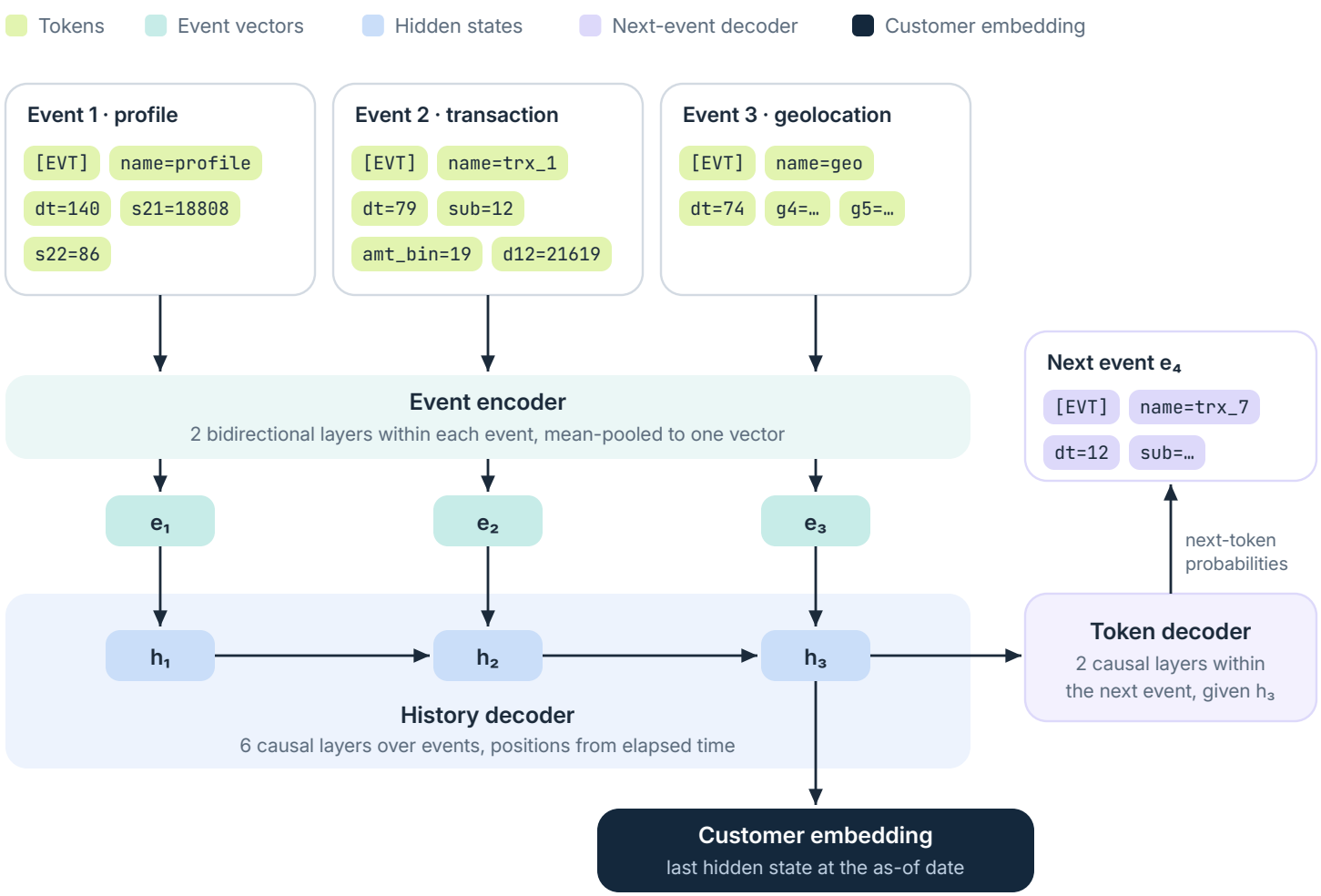
Tokens
Event vectors
Hidden states
Next-event decoder
Customer embedding
Event 1 · profile
[EVT]
name=profile
dt=140
s21=18808
s22=86
Event 2 · transaction
[EVT]
name=trx_1
dt=79
sub=12
amt_bin=19
d12=21619
Event 3 · geolocation
[EVT]
name=geo
dt=74
g4=…
g5=…
Event encoder
2 bidirectional layers within each event, mean-pooled to one vector
e1
e2
e3
h1
h2
h3
History decoder
6 causal layers over events, positions from elapsed time
Next event e4
[EVT]
name=trx_7
dt=12
sub=…
next-token probabilities
Token decoder
2 causal layers within the next event, given h3
Customer embedding
last hidden state at the as-of date